

Roll No.:

END-SEM EXAMINATION DECEMBER - 2024

Name of the Course: B.Tech.

Name of the Paper: Mathematical Foundations for Artificial Intelligence

Semester: III

Course Code: TCS 343

Time: 3 Hours

Maximum Marks: 100

Note:

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question.
- Total marks in each main question are twenty.
- Each question carries 10 marks.

Q1	(20Marks)	
(a)	Let $F: R^4 \rightarrow R^3$ be defined by $F(x, y, z, t) = (4x + 6y - 2z + 4t, 2x - 10y + 12t, 4y + 2z + 2t)$. Compute the matrix A for the given linear map with respect to the basis $B_1 = \{(1, 2, 3, 4), (1, 2, 3, 0), (1, 2, 0, 0), (1, 0, 0, 0)\}$ and $B_2 = \{(1, 2, 3), (1, 2, 0), (1, 0, 0)\}$.	CO1
(b)	Let W be the subspace of R^4 generated by the vectors $(1, -2, 5, -3), (2, 3, 1, -4)$ and $(3, 8, -3, -5)$. Find a basis and the dimension of W , and hence extend the basis of W to a basis of the whole space R^4 .	
(c)	Express the following system of equations into the matrix equation form $AX = B$: $4x - y + 6z = 16$; $x - 4y - 3z = -16$; $2x + 7y + 12z = 48$, $5x - 5y + 3z = 0$. Determine if this system of equations is consistent and if so find its solution.	
Q2	(20Marks)	
(a)	Find $\cos \theta$ for the angle θ between $A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$ in the vector space of 2×2 real matrices with inner product defined by $\langle A, B \rangle = \text{trace}(B^T A)$.	CO2
(b)	Find an orthonormal basis for the subspace U of R^4 spanned by $(1, 1, 1, 1), (1, 2, 3, 4), (1, -5, -3, -2)$.	
(c)	Let V be the vector space of all continuous functions on the interval $[a, b]$. The following defines a norm on V : $\ f\ = \int_a^b f(t) dt$. Give a geometrical description of $\ f\ $ and the distance $d(f, g)$.	
Q3	(20Marks)	
(a)	Perform the singular value decomposition of a matrix $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$.	CO3
(b)	Perform the eigen-decomposition and diagonalize the given matrix: $\begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$.	
(c)	Find the Cholesky decomposition for the following matrix: $\begin{bmatrix} 25 & 15 & -5 \\ 15 & 18 & 0 \\ -5 & 0 & 11 \end{bmatrix}$.	
Q4	(20Marks)	
(a)	Perform the linearization process (step-by-step) of $y^{1/2}x^{1/5}$ at the point $(1, 2)$.	CO4
(b)	Find the third-degree Taylor polynomial of $xe^y + 1$ at the point $(1, 0)$.	
(c)	Let $f(x, y, z) = y^2 e^{y^2 + x^2 + 3} \cos x$. Calculate the gradient of f at the point $(1, 1, 1)$ and calculate the directional derivative of f at the point $(1, 1, 1)$ in the direction of the vector $(1, -1, 2)$.	
Q5	(20 Marks)	
(a)	An automobile manufacturer has three factories: A, B, and C. They produce 50%, 30%, and 20% respectively, of a specific model of car. 30% of the cars produced in factory A are white, 40% of those produced in factory B are white, and 25% produced in factory C are white.	CO5

	i. If an automobile produced by the company is selected at random, find the probability that it is white. ii. Given that an automobile selected at random is white, find the probability that it came from factory B.	
(b)	An airline operates between Dehradun and New Delhi. The flight time may be modelled by a normal distribution with a mean of 85 minutes and standard deviation 8. In order to boost sales of the service, the airline decides to refund the fares if a flight time exceeds the mean flight time by t minutes. The airline does not want to refund more than 0.005 of the fares. Find the value of t , correct to the nearest minute.	CO6
(c)	Two manufacturers supply blankets to emergency relief organizations. Manufacturer A supplies 3000 blankets and 4% are irregular in workmanship. Manufacturer B supplies 2400 blankets and 7% are found to be irregular. Given that a blanket is irregular, find the probability that it came from manufacturer B.	